

NG_transmission_emissions_r1.R

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Transmission	<i>Create gridded natural gas transmission methane emissions maps</i>
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Description

‘Transmission’ writes 2 netcdf files of gridded methane emissions from natural gas transmission, as well as optional visuals and 2 optional csv files.

Usage

```
Transmission(  
  input_directory,  
  GHGI_transmission_compressors,  
  GHGI_Pipeline,  
  HIFLD_compressor_file,  
  domain,  
  domain_template,  
  ghgrp_facility_info,  
  state_name_list,  
  output_directory,  
  inventory_year,  
  verbose,  
  plot_directory,  
  County_Tigerlines,  
  State_Tigerlines  
)
```

Arguments

input_directory	Character providing the full filepath to save/load raw input data
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GHGI_transmission_compressors	Character or data.frame. Pulled from config file. Either GHGI to indicate the GHGI file should be used to pull emissions and activity data or a data frame providing the needed values.
GHGI_Pipeline	Character or data.frame. Pulled from config file. Either GHGI to indicate the GHGI file should be used to pull emissions and activity data or a data frame providing the needed values.
HIFLD_compressor_file	Character providing the full filepath to the HIFLD compressor data. As this file is now deprecated and no replacement has been created, it currently must be provided as part of the package.
domain	SpatVector polygon outlining the desired output area
domain_template	SpatRaster providing the desired output grid, including the desired resolution and coordinate reference system
ghgrp_facility_info	Data.frame with the GHGRP location data for all years and states. See https://www.epa.gov/enviro/en-data-service-api
state_name_list	Character vector listing all states within the desired domain
output_directory	Character providing the full filepath to save processed data
inventory_year	Character indicating the desired year of data to use.
verbose	Logical indicating whether to save additional output. This includes plots of the gridded methane emissions for each fuel-sector-inventory-variation combination as well as 2 summed plots for each inventory-variation combination - one for wood and one for all other sectors.
plot_directory	Character providing the full filepath to save figures. Only relevant if verbose = TRUE.
County_Tigerlines	SpatVector. United States Census Bureau county shapefile downloaded in Main.
State_Tigerlines	SpatVector. United States Census Bureau county shapefile. Available at https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html . Only relevant if verbose=TRUE.

Details

This function calculates and grids methane emissions from natural gas transmission systems. It uses the Homeland Infrastructure Foundation-Level Data (HIFLD), Environmental Protection Agency's (EPA) Greenhouse Gas Inventory (GHGI), EPA Greenhouse Gas Reporting Program (GHGRP) and Energy Information Administration (EIA) Energy Atlas data.

The necessary GHGRP, GHGI, and EIA data will be automatically downloaded.

For pipelines the GHGI data includes the emissions and activity data for pipeline leaks, meter and regulating stations, and venting. An emission factor in mols of methane per meter of pipeline per second can then be calculated. This emission factor is then applied to the EIA inter/intrastate pipeline location data.

For compressors the GHGI data includes the emissions and activity data for compressor station fugitive emissions, dehydrator vents, flaring, engines, turbines, generators, pneumatic devices, and

venting. For generators they are split into engine and turbine emissions so the ratio between transmission/storage emissions for engines and turbines are applied to the generator values to get the transmission component of generator emissions. An national average emission rate in mols of methane per station per second can then be calculated. This emission rate is then assigned to all HIFLD compressors. GHGRP data is then used to overwrite this default emission rate. Note most compressor stations do not report their emissions to the GHGRP, so only a subset will be overwritten using GHGRP data. The GHGRP compressor emissions are scaled so that the average emissions within the domain are equal to the national average calculated from the GHGI. As there is not a common identifier between the GHGRP and HIFLD datasets, the nearest facility is considered the matching facility.

The GHGRP includes only facilities that emit at least 25,000 metric tons of carbon dioxide equivalent while the GHGI is intended to capture all national emissions. GHGRP data is available starting in 2010 and generally is about 2 years behind present day, the GHGI is available starting in 1990 and is updated approximately in sync with the GHGRP. Both datasets are annual. The GHGRP is at the facility scale while the GHGI is national totals.

The HIFLD compressor data being used was last updated December 2022. It has since been deprecated, but as there is no replacement available, it is still being used.

The GHGI is available at <https://www.epa.gov/ghgemissions/inventory-us-greenhouse-gas-emissions-and-sinks-1990-2022>. The necessary GHGI Annex data is available at <https://www.epa.gov/ghgemissions/natural-gas-and-petroleum-gas-emissions-1990-2022> for the 2024 GHGI. In the GHGI Annexes, available at <https://www.epa.gov/ghgemissions/inventory-us-greenhouse-gas-emissions-and-sinks-1990-2022>, for 2024 there is a link to the file in Section 3.6: "Methodology for Estimating CH₄, CO₂, and N₂O Emissions from Natural Gas Systems". The excel file has multiple sheets, each of which has a separate layout. The GHGRP is available at <https://ghgdata.epa.gov/ghgp/main.do>, and the EIA data is available at <https://atlas.eia.gov/datasets/eia:natural-gas-interstate-and-intrastate-pipelines/about>.

Value

Nothing is returned from the function, but the main outputs are 2 netcdf files of the methane emissions from natural gas transmission. They are titled "NG_trans_compressors.nc" and "NG_trans_pipes.nc" where NG=natural gas and trans=transmission. The first file is for transmission compressor emissions and the second is for transmission pipeline emissions.

2 csv files are also optionally saved. These are "NG_trans_compressors.csv" and "NG_trans_compressors_all.csv". The simpler csv includes only the name, location, and assigned emissions for compressors within the domain that were pulled from the GHGRP. The _all files include all variables that were in the corresponding input file for the same compressors.

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Examples

```
library(terra)
grid_bbox=cbind(c(-76.65,-73.65),c(38.97,40.97))
grid_res=0.01
grid_crs="epsg:4326"
grid <- rast(nrows=diff(range(grid_bbox[,2]))/grid_res,
             ncols=diff(range(grid_bbox[,1]))/grid_res, xmin=min(grid_bbox[,1]),
             xmax=max(grid_bbox[,1]), ymin=min(grid_bbox[,2]), ymax=max(grid_bbox[,2]),
```

```
      crs=grid_crs)
grid_vect <- as.polygons(ext(grid),crs=grid_crs)
Transmission(GHGI_transmission_compressors="GHGI",
             GHGI_Pipeline="GHGI",
             HIFLD_compressor_file="~/../Desktop/in/Natural_Gas_Compressor_Stations.csv",
             ghgrp_facility_info="~/../Desktop/in/GHGRP/facility_info.csv",
             domain=grid_vect,
             domain_template=grid,
             state_name_list=c("DE","MD","NJ","NY","PA"),
             output_directory="~/../Desktop/out/",
             input_directory="~/../Desktop/in/",
             inventory_year=2018,
             verbose=TRUE,
             State_Tigerlines=vect("~/../Desktop/in/State_Tigerlines/tl_2018_us_state.shp"),
             County_Tigerlines=vect("~/../Desktop/in/County_Tigerlines/tl_2018_us_county.shp"),
             plot_directory="~/../Desktop/plots/")
```

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